

ETHAN (E) H. HARRISON

ehharrison13@gmail.com ◇ (310) 947-1395 ◇ [linkedin.com/in/ehharrison13/](https://www.linkedin.com/in/ehharrison13/)

ACADEMICS

University of California, Berkeley

Bachelor of Science, Electrical Engineering and Computer Science

August 2023 - Present

Current GPA: 3.97/4.00

Relevant Coursework

- CS188: Introduction to Artificial Intelligence (A)
- CS189: Introduction to Machine Learning (A)
- EECS182: Designing, Visualizing and Understanding Deep Neural Networks (A)
- CS185: Deep Reinforcement Learning, Decision Making, and Control (A)
- CS170: Efficient Algorithms and Intractable Problems (A)
- EECS106A: Introduction to Robotics (A+)
- EECS106B: Robotic Manipulation and Interaction (A)
- EECS127: Optimization Models in Engineering (A+)

WORK AND RESEARCH EXPERIENCE

Berkeley Robotic AI & Learning Lab

Research Advised Under Professor Sergey Levine and Postdoc Andrew Wagenmaker

February 2025 - Present

Berkeley, CA

- Collaborated with the Siemens' Robotic Research Lab to develop improved methods of finetuning generalist robot policies under inference delay assumptions, with paper submitted to CoRL 2026 (Decision pending)
- Collected over 300k timesteps using OpenVLA and π_0 models on real-world environments in order to train RL algorithms for the real world
- Achieved state-of-the-art performance for offline RL algorithms to manipulate real-world environments, getting a 30% improvement over BC for the same tasks using IQL and SAC+BC

Northrop Grumman Systems Engineering and Integration

Intern for Advanced Responsive Integrated Systems (ARIS)

Summers 2022 - 2024

Space Park, CA

- Developed the Requirement Verification Revision burndown planner for scheduling and verifying system requirements under the Requirements and Yield Output (RAYO) software, successfully integrating with mission data
- Rebuilt the Mission Database platform for ARIS in the current RAYO architecture, greatly improving the scalability and maintainability of the application
- Modeled new part creation equipment for the Modular Bi-axis Assembly in the Mechanisms and Deployable Products department and received an Excellent rating for the end-of-internship performance review
- Implemented time-series forecasting models to predict employee attrition within the software systems department, presenting to high-level managers and achieving division-wide deployment

UC Berkeley Undergraduate Course Staff

CS 61A Teaching Assistant in the Structure and Interpretation of Computer Programs

January 2024 - Present

Berkeley, CA

- Managed office hours for a class of 1,500 students in the fundamentals of computer science, contributing to a 7% increase in student test scores and a 12% increase in student satisfaction for the class

PAPERS

Learning to Act While Waiting: RL Finetuning of Generalist Robot Policies Under Inference Latency

Submitted to Conference on Robot Learning (CoRL) 2026

- Investigated the ability to finetune action chunking VLA policies when considering inference time constraints
- Proposed a new method, Asynchronous RL with Intermediate Information (ARLI), which achieves state-of-the-art performance on both simulated and real-world manipulation tasks

PROJECTS

REAL Real Steel: Human-to-Robot Teleoperation and Punch Planning

EECS106A Introduction to Robotics, Best Final Project ([Link](#))

- Engineered a physical harness system with IMUs and an ESP32 to allow users to teleoperate a UR7e robot using just arm movements, capable of pick-and-place tasks and other dexterous movements
- Included computer vision using MediaPipe and ArUco detection for hand movements and improved teleop accuracy
- Designed a GUI using ROS and RViz to allow users to define and execute robot punches

Learning to Race: Deep Reinforcement Learning for Autonomous Drone Navigation

EECS106B Robotic Manipulation and Interaction, Final Project ([Link](#))

- Trained and evaluated a PPO-based drone racing policy with custom reward and termination design in Isaac Sim to fly a quadrotor through a multi-gate race course as quickly as possible
- Incorporated parallel simulation environments and curriculum-style reset tools to improve the rate of policy convergence even with limited computational power
- Achieved the fastest policy improvement rate among all other teams and achieved the fastest average lap completion for teams using RL-based strategies

Sah-AI: WandB Experiment-Management Agent With Tool Calling

EE194-16 Scalable AI: Inference and Serving Project ([Link](#))

- Created an intractable, agentic AI system that designs and executes machine learning experiments in WandB based on the user's codebase, project, and request.
- Developed the tool-calling backend using OpenAI's tool API to allow the agent to interact with WandB, read and write to the user's codebase, and summarize experimental results
- Recorded system performance and quality metrics to determine the best serving backend and model system prompts

EXTRACIRRICULARS

Eta Kappa Nu Electrical and Computer Engineering Honor Society

January 2026 - Present

Executive, Department Relations Officer

Berkeley, CA

- Organized the creation and distribution of the 2026 EECS Undergraduate Survey, reaching 80% more students than last year's survey and presenting the survey's results to the entire UC Berkeley EECS faculty
- Held bi-weekly meetings with the chair of the UC Berkeley EECS Department, Jelani Nelson, to discuss the current state desires of EECS student organizations
- Provided free tours of the UC Berkeley EECS Department for high school students and hosted the EECS Student Panel for incoming EECS freshmen

ANova - Computer Science Education and Mentoring

September 2023 - Present

Officer of Curriculum, General Meetings Staff

Berkeley, CA

- Volunteered for weekly computer science lessons at 10+ middle and high schools for 200+ students around the East Bay, prioritizing students and schools in under-resourced communities
- Led the Curriculum committee in Anova, working with 9 other students to help develop and improve upon ANova's current curriculum, transitioning the club to a new platform for labs and projects
- Hosted workshops and provided one-on-one sessions with high school seniors to help them in the college application process

Berkeley Model United Nations (BMUN)

September 2023 - Present

Tech Deputy for External Affairs, UN Office of Drugs and Crime Chair

Berkeley, CA

- Organized Berkeley's annual High School Model United Nations conference, successfully holding the conference for over 2000 students from across the country
- Managed and updated BMUN's website, Huxley, creating an AI-detection software for position papers submitted to the conference

SKILLS

Software	C, Java, Python, SQL, ROS, Javascript, HTML, Git
Frameworks	Django, Pandas, Numpy, PyTorch, Tensorflow, Jax/Flax
Hardware	GD&T, Circuit Design/Analysis, Electronics Design/Prototyping